

**Department of Artificial Intelligence and Data Science****Vision of the Department**

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission of the Department

To nurture computer engineering professionals by offering:

- *Dynamic learning environment that encourages problem-solving using emerging technology*
- *National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry-institute collaboration*
- *Holistic, ethical and sustainable development for lifelong success.*

Session 2026-2027

Vision: Dream of where you want.	Mission: Means to achieve Vision
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Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	Preparation	P: Preparation	Pep-CL abbreviation pronounce as Pep-si-IL easy to recall
PEO2	Core Competence	E: Environment (Learning Environment)	
PEO3	Breadth	P: Professionalism	
PEO4	Professionalism	C: Core Competence	
PEO5	Learning Environment	L: Breadth (Learning in diverse areas)	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:

Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

“I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life.” to contribute to the development of cutting-edge technologies and Research.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Krishna Mishra

Name and Signature of Student and Date

(Signature and Date in Handwritten)

15/07/2026



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Session	2025-26 (EVEN)	Course Name	LAB: Optimization Techniques
Semester	7th	Course Code	23ADS1707
Roll No	41	Name of Student	Krishna Mishra

Practical Number	05
Course Outcome	CO1: Understand unconstrained and constrained optimization techniques. CO2: Apply non-linear optimization techniques to solve engineering problems. CO3: Analyze optimization problems using operations research techniques. CO4: Evaluate transportation and transshipment optimization problems.
Aim	To solve a constrained optimization problem involving inequality constraints using the Kuhn-Tucker (KKT) Conditions.
Problem Definition	Optimization Using Kuhn-Tucker (KKT) Conditions
Theory (100 words)	This program uses the KKT (Karush-Kuhn-Tucker) conditions to solve a constrained optimization problem. It minimizes the objective function $f(x)=x^2$ subject to the constraint $x \geq 2$. A Lagrangian function is formed using a multiplier λ . Stationarity and complementary slackness conditions are applied to find the solution. The optimal solution is $x=2$, giving a minimum value of 4 .
Procedure and Execution (100 Words)	1. Define the variable x and Lagrange multiplier λ. 2. Define the objective function $f(x)=x^2$ and constraint $x \geq 2$. 3. Construct the Lagrangian function using the objective and constraint. 4. Calculate the stationarity condition by differentiating the Lagrangian with respect to x. 5. Apply the complementary slackness condition and solve for x and λ. 6. Check the feasibility conditions $x \geq 2$ and $\lambda \geq 0$. 7. Calculate the minimum value of the objective function. 8. Display the optimal solution and minimum value.



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Code:

```
import sympy as sp

# Define variables
x, lam = sp.symbols('x lam', nonnegative=True)

# Objective and constraint
f = x**2
g = 2 - x

# Lagrangian
L = f + lam * g

# KKT conditions
eq1 = sp.diff(L, x)
eq2 = lam * g

# Solve
solutions = sp.solve((eq1, eq2), (x, lam), dict=True)

print("Possible Solutions:")
print(solutions)

for sol in solutions:
    x_val = sol[x]
    lam_val = sol[lam]

    if x_val >= 2 and lam_val >= 0:
        print("\nOptimal Solution")
        print("x =", x_val)
        print("Lambda =", lam_val)
        print("Minimum Value =", f.subs(x, x_val))
```

Output:

Possible Solutions:
[{lam: 0, x: 0}, {lam: 4, x: 2}]

Optimal Solution

x = 2

Lambda = 4 Minimum Value = 4



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Output Analysis	The program gives the optimal solution as $x = 2$ and $\lambda = 4$. The constraint $x \geq 2$ is satisfied. The minimum value of the objective function x^2 is 4. Therefore, the optimal solution is $x = 2$ with minimum value 4.
Link of student Github profile where lab assignment has been uploaded	krish-na45/Optimization-Technique-Practicals
Conclusion	The optimal point is $((5,5))$, where the objective function attains its minimum value of 50.
Plag Report (Similarity index < 12%)	4% Plagiarized Exact Match 4% Partial Match 0%
Date	15/07/2026